Assignment No: 6

Title: Create a Docker container for an AI model and test it on a local machine to ensure consistent performance across environments

Steps:

Phase 1: Prepare the AI Application

First, you need to set up the local files that will be packaged into the container:

1. **Create a Project Directory**: Open your terminal and create a folder to house your project.

```
None
mkdir ai_docker_project
cd ai_docker_project
```

2. **Create the Python Script**: Create a file named model.py and add the inference logic.

```
None
import numpy as np
print("AI Model Output:", np.mean([10, 20, 30, 40]))
```

3. **Define Dependencies**: Create a requirements.txt file and list the numpy library. Plaintext

```
None
numpy
```

Phase 2: Create the Dockerfile

The Dockerfile serves as the blueprint for your environment:

4. **Write the Dockerfile**: Create a file named dockerfile (no extension) in the same directory and add these instructions:
    - **Base Image**: Use python:3.10-slim.
    - **Working Directory**: Set it to /app.
    - **Copy Requirements**: Move the dependency list into the container.
    - **Install Dependencies**: Run pip install to set up the environment. ○ **Copy Script**: Move your model.py into the container.
    - **Set Command**: Define the command to run the model.

        (file content given at end of this docs)

Phase 3: Build and Test

Now, convert your blueprint into a usable image and run it:

5. **Build the Docker Image**: Run the build command in your terminal.

```
None
```

```
docker build -t ai-model-container .
```

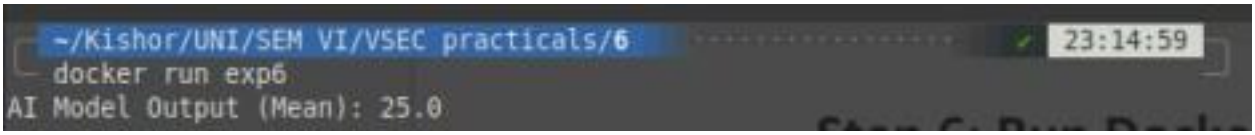
6. **Run the Container**: Execute the model within the isolated environment.

```
None
```

```
docker run ai-model-container
```

7. **Verify Output**: Confirm that the output (25.0) matches what you would expect from a local execution.

Output:



Dockerfile

None

```
# Use a slim Python image to keep the container lightweight FROM python:3.10-slim

# Set the working directory inside the container
WORKDIR /app

# Copy the requirements file and install dependencies COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy the model script into the container COPY model.py .

# Define the command to run the script when the container starts
CMD ["python", "model.py"]
```